

AUTO TICKET CLASSIFICATION USING FLOW DESIGNER

PHASE 1: BRAINSTORMING AND IDEATION

1. Project Overview & Institutional Background

In modern educational ecosystems, information technology forms the backbone of day-to-day academic and administrative workflows. From interactive smart projectors in lecture halls to campus-wide high-density Wi-Fi networks and centralized portal identity systems, faculty and students rely continuously on digital infrastructure.

The institutional IT Helpdesk operates as the single point of contact (SPOC) for reporting and resolving technical anomalies. However, as student enrollments increase and technology adoption accelerates, helpdesk operations face severe strain. This project, titled 'Auto Ticket Classification using Flow Designer', introduces an enterprise-grade automated triage mechanism within the ServiceNow platform to streamline incident lifecycle management from the point of entry.

- **Project Scope:** End-to-end automation of incident intake, parsing, classification, dependent subcategorization, and initial caller communication.
- **Primary Technology:** ServiceNow Flow Designer (Process Automation Workflow Studio).
- **Target Environment:** School & University IT Helpdesk operations.
- **Deployment Methodology:** Global Update Set architecture with zero script dependencies.

2. Current Operational Bottlenecks & Problem Statement

At present, the school IT helpdesk receives a high volume of unstructured technical requests daily from across multiple campus departments and academic blocks. These requests encompass recurring

operational issues, including Wi-Fi connectivity degradation, classroom projector bulb/HDMI failures, forgotten portal credentials, and sluggish computer laboratory workstations.

Currently, IT staff must manually open each logged incident, read through subjective descriptions, manually select the appropriate parent category, identify the corresponding subcategory, and route the ticket to resolver teams. This legacy approach suffers from several critical operational defects:

- **High Initial Response Latency:** Tickets frequently remain sitting in unassigned triage queues for 2 to 4 hours before manual review takes place.
- **Inconsistent Categorization:** Different service desk technicians apply varying category tags to identical problems, skewing analytics and service metrics.
- **High Operational Triage Overhead:** Up to 25% of Tier-1 technician working hours are consumed solely by sorting and re-assigning incoming tickets rather than active troubleshooting.
- **Lack of Instant User Confirmation:** End users (students and faculty) receive delayed acknowledgements, leading to duplicate ticket submissions for the same issue.

3. Brainstorming & Architectural Solution Exploration

During the initial ideation sprint, the project team conducted structured brainstorming sessions to evaluate potential technical architectures supported by the ServiceNow platform. The goal was to establish a solution that is robust, immediately maintainable, highly scalable, and capable of eliminating manual classification overhead.

Approach Evaluated	Technical Mechanism	Key Advantages	Disadvantages & Risks	Verdict
Approach A: Business Rules	Before/After DB Insert script using GlideRecord JavaScript	Direct database trigger, ultra-low latency	Requires server-side scripting, high maintenance debt, hard for junior admins to alter	Rejected
Approach B: Predictive Intelligence	Machine Learning models trained on historical ticket text	Handles complex NLP and sentiment analysis	Requires thousands of historical training records, costly licensing, risk of model drift	Rejected
Approach C: Flow Designer Automation	Triggered workflow engine with visual rule conditions and actions	100% no-code, drag-and-drop rule maintenance, real-time logging, native email actions	Requires predefined keyword rules in flow logic	Selected & Approved

4. Project Objectives & Success Criteria

The ideation phase finalized four fundamental project objectives designed to deliver measurable business and academic value:

- **Objective 1 — Real-Time Automated Classification:** Achieve instantaneous population of both Category and Subcategory attributes upon record creation.
- **Objective 2 — Radical Reduction of Manual Workload:** Decrease manual ticket categorization effort by 100% for standard recurring classroom and laboratory incidents.

- **Objective 3 — Automated Omnichannel Notification:** Dispatch immediate automated acknowledgment notifications with dynamic incident references to callers.
- **Objective 4 — Maintainability & No-Code Scalability:** Deliver an extensible system architecture that enables IT administrators to introduce new issue categories without writing a single line of JavaScript.